Happy Supplies Parts Warehouse

original table

customerNumber	customerName	CustomerType	**date**	**time**	employee	**partNumber**	name	type	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

The original table is already in 1NF
It contains primary keys: customer number, date, time, part number
There is no replication in the table

2NF
We can see that there is partial dependency in the table, some attributes only depend on part of the primary key, not all of them.
To be more specifically:
(1) customerName, customerType only depend on CustomerNumber
(2) partName, partType, cageCode, unitPrice only depend on partNumber
(3) employee only depends on customerNumber, date, and time

customer table

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

part table

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

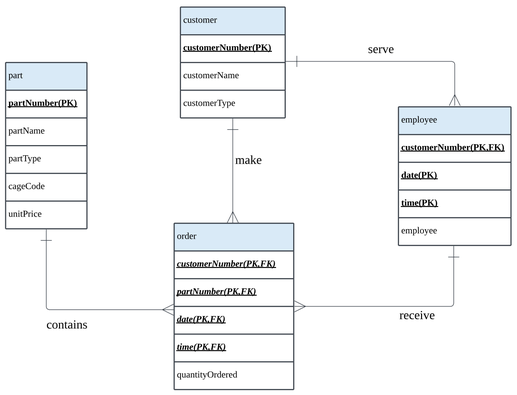
employee table

customerNumber	**date**	**time**	employee
HG54587	7/1/2024	10:30 am	D. Harrison

order table

customerNumber	**date**	**time**	**partNumber**	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

3NF
since there is no transitive dependency in each table, the data are already in 3NF



Panacea Mental Health Corporation

original table

staffNo	therapistName	patNo	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

it is already in 1NF since it has composite primary keys and there's no replication

2NF
there is partial dependency in this case
(1) therapistName only depend on staffNo
(2) branchNo only depend on staffNo and appointmentDate

staff table

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

branch table

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment table

staffNo	**appointmentDate**	**appointmentTime**	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

3NF
in patient table, it is clear that patient name only depend on patient number

patient table

patNo	patName
P100	Lily
P105	Jill Baker
P108	Andy
P110	Jimmy

appointment table

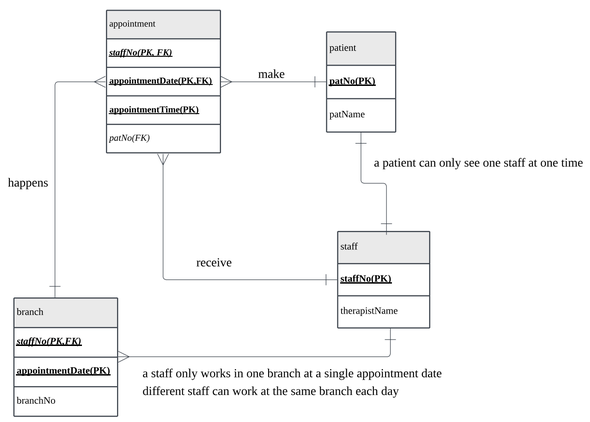
staffNo	**appointmentDate**	**appointmentTime**	patNo
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

staff table (unchanged)

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

branch table (unchanged)

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13



Maid Better

original table

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

it is already in 1NF since is has primary keys and there's no replication

2NF
(1) eName only depend on eNo
(2) eventNo and eventLoc only depend on contractNo

employee table

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

contract table

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

work_assignment table

eNo	***contractNo***	hours
1135	*C1024*	16
1057	*C1024*	24
1068	*C1025*	28
1135	*C1025*	15
1135	*C1026*	10

3NF
eventLoc only depend on eventNo

event table

eventNo	eventLoc
H25	Queens
H4	Yonkers
1068	White T

contract table

contractNo	*eventNo*
C1024	*H25*
C1025	*H4*
C1026	*H25*

employee table (unchanged)

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

work_assignment table (unchanged)

eNo	***contractNo***	hours
1135	*C1024*	16
1057	*C1024*	24
1068	*C1025*	28
1135	*C1025*	15
1135	*C1026*	10

